have almost zero startup time, and add a premium of just $1-2 per device. Such a device would provide localized haptic feedback which would let you feel, for example, a ball bouncing and striking the sides of a screen. In real applications, this could be even used to provide specific feedback for certain reminders, or an email from an important contact. And of course, the possibilities are endless as far as games are concerned.

While actuators use vibrations, another way that researchers have been trying to provide haptic feedback is by creating textures on the screen. Imagine feeling a button on screen as you run your hand over it. One company that has already demonstrated such a product is Senseg. Founded in 2008, the company has created a tactile technology that uses electrostatic fields to exert different levels of force on the fingertip, and thus simulates the sensation produced by different surfaces. So, when the devices debut, it would allow you to feel and recognise everything from sand to bubble wrap just by touching the screen. While Senseg's technology will work with all the current panels, there is another company that is working on special screen to create the same virtual texture.

Senseg's haptic feedback

In early May this year, an Apple patent emerged that proposes the use of multi-tiered elastic screen to create raised buttons and other identifiable textures and patterns. While Senseg's technology just simulates these textures, Apple's patent proposes actual creation of these crests and troughs using flexible OLED. It remains to be seen what version of iPhone/iPad will actually see this tech, but the possibilities are extremely exciting. It might also put to rest the eternal debate over physical keyboards.